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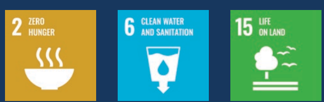
SUPPORT FOR THE ESTABLISHMENT OF MODEL GUIDE FIELDS IN SUGARCANE GROWING IN UPPER EGYPT

March 2026

SDGs:



Country:	Egypt
Project Code:	TCP/EGY/3902
FAO Contribution:	USD 350 000
Duration:	31 July 2023–31 December 2025
Contact Info:	FAO Representation in Egypt FAO-EGY@fao.org

Implementing Partners

Agricultural Research Center (ARC), Ministry of Agriculture and Land Reclamation (MALR).

Beneficiaries

Smallholder farmers; farmer field school (FFS) facilitators; master trainers; training centres; local communities; agricultural institutions; private sector partners; government authorities; technical experts.

FAO Programme Priority Area(s)

Better Production (BP): 1.

Country Programming Framework (CPF) Outputs

CPF 2023–2027

Output 2.4.1: Research and development, investment programming and multiagency partnerships supported to transform selected agrifood systems.

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Background

Sugarcane is a perennial subtropical crop and represents the most important agricultural commodity in Upper Egypt, where it is cultivated along the fertile banks of the Nile River. The climatic conditions of the region, characterized by high temperatures, strong light intensity and dry weather, are favourable for sugarcane growth and sugar accumulation. Cultivation relies almost entirely on conventional flood irrigation using water from the Nile.

Sugarcane has high socioeconomic importance in Upper Egypt, which is among the poorer regions of the country. According to Sugar Crop Council reports in 2021, the total cultivated area is about 141 700 hectares (340 000 faddan), of which approximately 102 000 hectares (243 000 faddan) are devoted to sugar production. The remaining area is used for seed cane, molasses, chewing cane and juice production. Production is largely dependent on small-scale farmers owning less than one hectare. The sector provides a major source of income for more than 150 000 farmers and about 21 000 workers employed by the Egyptian Sugar and Integrated Industries Company (ESIIC).

Owing to its economic and social importance, the Government of Egypt places high priority on improving sugarcane production and productivity.

Cane pricing is based on weight rather than sucrose content, encouraging biomass production over sugar yield and driving excessive irrigation and nitrogen use. Water scarcity is a major constraint, with national policy prioritizing drinking and industrial use, leaving agriculture with limited allocation. Studies indicate sugarcane requires 8 450–12 000 m^3 per feddan annually, with peak demand from June to August.

The Updated Strategy for Sustainable Agricultural Development 2030 aimed to increase productivity to 55 tonnes/feddan, reduce water and mineral fertilizer use by 25 percent, and raise total sugar production. Measures included modernizing cultivation through transplanting and drip irrigation. Pre-sprouted single-bud seedlings reduce seed cane requirements by up to 80 percent, improve plant density, facilitate intercropping and support mechanization. Drip irrigation reduced water use by approximately 30 percent.

Despite these measures, significant structural and technical constraints continue to affect sugarcane production. The prevailing planting method produces dense and tangled stands that hinder effective irrigation management and restrict the use of mechanization. High seed cane requirements and the absence of systematic quality control reduce planting efficiency. Inadequate land levelling and poor alignment with field slopes limit water distribution uniformity. Irrigation scheduling is generally based on fixed shifts rather than crop growth stages, leading to over- or under-irrigation. Continued reliance on flood irrigation, combined with farmers' reluctance to adopt modern systems due to concerns about water charges, contributes to excessive water use. Flood irrigation is also associated with high nitrogen fertilizer application rates, resulting in low nutrient-use efficiency. Furthermore, the supply of sugarcane to factories is based on weight rather than sugar content, reinforcing incentives for biomass production at the expense of quality.

In response to these challenges, the Ministry of Agriculture and Land Reclamation (MALR) requested technical assistance from the Food and Agriculture Organization of the United Nations (FAO) to support the modernization of sugarcane production systems. This project sought to strengthen sustainable sugarcane production in Upper Egypt by promoting modern technologies, improving water and input management and enhancing farmer livelihoods, while contributing to national food security and long-term resource conservation.

IMPACT

The project made a measurable contribution to enhancing sugarcane productivity and improving smallholder livelihoods in Upper Egypt, reaching 500 farmers managing 594 feddans. Through the systematic application of climate-smart agriculture (CSA) practices within farmer field schools (FFSs) and demonstration plots, farmers adopted efficient sugarcane production techniques that reduced input requirements, optimized yields and supported sustainable management of natural resources, including water, land and fertilizers.

Preliminary observations from the 20 FFS demonstration fields indicate that transplanted sugarcane grown on raised beds, combined with timely and uniform earthing up, optimized irrigation scheduling, improved fertilization and strategic intercropping, substantially enhanced crop performance. Expert field inspections and sampling showed seven to 15 tillers per plant, confirming the positive effect of earthing up on basal tillering and stalk density, which directly correlates with yield potential. Estimated yields ranged from 45 to 65 tonnes per feddan, averaging approximately 55 tonnes per feddan, representing a 62 percent increase over the baseline of 34 tonnes per feddan. This far exceeds the end-of-project target of a 10 percent yield increase, demonstrating the agronomic effectiveness of the promoted CSA practices at the demonstration level.

Sustainability and broader impact will depend on continued adoption and replication beyond FFS plots. Qualitative evidence from focus group discussions and farmer feedback indicates strong willingness among participants to continue applying the practices independently, highlighting the potential for farmer-to-farmer diffusion in neighbouring villages. The project also contributed to national and international development priorities, supporting the Sustainable Development Goals (SDGs) on poverty reduction, food security, water-use efficiency and sustainable economic growth, while reinforcing national strategies for climate-smart sugarcane cultivation. Follow-up actions include a comprehensive post-project survey to quantify yield gains, assess resource-use efficiency, and document farmer success stories to support scaling up and long-term impact.

ACHIEVEMENT OF RESULTS

The project effectively addressed key challenges in sugarcane production in Aswan, including low productivity, inefficient water use, inadequate pest and nutrient management and limited adoption of CSA practices.

The FFSs proved highly relevant to local needs, and the project design ensured results-based management through clearly defined objectives, outcomes and outputs, supported by a comprehensive results framework, baseline data, performance indicators and aligned resource allocation. The introduction of innovations such as raised-bed transplanting, optimized irrigation scheduling, earthing up to promote basal tillering, precise fertilization, integrated pest management (IPM) and intercropping enabled farmers to improve yields, resource efficiency and overall crop management. Stakeholders demonstrated strong commitment to supporting replication of these practices in surrounding villages, highlighting the project’s potential for scaling and sustainability.

More specifically, under Output 1, the project partially operationalized two training centres linked to sugarcane transplant production nurseries in Kom Ombo and Edfu. The Kom Ombo centre was fully functional, associated with a 40 feddan demonstration and seedling multiplication area, and served as a multipurpose hub for classroom instruction, practical training, facilitator mentoring, FFS planning, field observation review, data collection and reporting. The Edfu centre remained under construction and could not be rehabilitated during the project implementation. To mitigate this, a five feddan demonstration field in Wadi El-Sa‘idia (Al-Iman Village) provided a decentralized training platform for hands-on learning, practical demonstration of raised-bed techniques and intercropping. Training tools, furniture and equipment were identified, procured, installed and made operational at Kom Ombo, supporting technical sessions, meetings, reporting and monitoring. Demonstration plots at both sites effectively showcased transplanted sugarcane, raised-bed cultivation, intercropping and irrigation management, reinforcing productivity and resource-use efficiency messages. Follow-up actions include completing the Edfu facility, formal handover, integration into national programmes and maintenance of demonstration plots as reference sites.

Under Output 2, the capacities of national cadres as master trainers were strengthened through a training of trainers (ToT) programme. Twenty-two national cadres, exceeding the target of 20, were selected based on technical expertise, extension experience and institutional roles, representing extension services, the Sugar Crops Research Institute, sugar factories, universities and technical schools, with 13.6 percent female participation. The ToT, held in Aswan, combined CSA principles, modern sugarcane production practices and FFS facilitation skills, achieving pre- to post-training knowledge gains from 45.6 percent to 99.4 percent.

Master trainers subsequently supported 20 FFSs, providing technical guidance and building farmer confidence. Follow-up actions include refresher trainings, continued engagement in future FFS cycles, and integration into national extension programmes.

Under Output 3, the capacities of FFS facilitators were enhanced through the training of 47 participants, surpassing the target of 20. Facilitators received comprehensive instruction covering CSA, water management, facilitation skills, agricultural extension and socioeconomic analysis (AESA), planning and adult learning. Hands-on field exercises and peer learning strengthened applied skills, confidence and consistency despite differences in experience. Draft CSA curricula for sugarcane FFSs were developed and refined based on farmer feedback and field realities. Trainers were selected from FAO experts, the Sugar Crops Research Institute, and experienced FFS practitioners to address the limited pool of certified national trainers. Five-day facilitator trainings combined classroom, group and field-based practice, bridging technical knowledge with facilitation skills. Follow-up includes on-the-job coaching, structured peer learning and periodic refresher sessions.

Under Output 4, 20 FFS demonstration fields were established across Kom Ombo and Edfu, engaging 500 farmers managing 594 feddans. The fields introduced transplanted sugarcane on raised beds, earthing up, intercropping, optimized irrigation, nutrient management and pest control. Raised beds allowed irrigation in two hours versus twelve on flat beds, improved basal tillering, reduced waterlogging and enhanced labour efficiency. Farmers reported healthier plants, easier field management and strong willingness to adopt practices independently. Continuous technical backstopping to facilitators ensured high-quality implementation across all fields, focusing on irrigation scheduling, fertilizer and nutrient management, transplanted sugarcane care, intercropping, pest and disease management, and participatory facilitation. All FFS fields were equipped with essential tools and inputs, including row markers, earthing up implements, calibrated fertilizer application tools, transplants, fertilizers, micronutrients, humic substances and knapsack sprayers. Twenty structured field days enabled farmers to observe and compare traditional and CSA practices at key growth stages, strengthening analytical skills, decision making and confidence. Follow-up actions include monitoring adoption, assessing resource-use efficiency and documenting farmer-led replication.

Under Output 5, a participatory monitoring, evaluation and networking framework supported all 20 FFSs. Facilitators systematically documented activities using diaries, observation checklists, reflection sessions, and learning plots to track performance and adoption of practices. Digital nursery-level databases were established at Kom Ombo and Edfu, enabling real-time data aggregation, cross-site analysis and evidence-based reporting. Continuous backstopping by master trainers and project staff enhanced facilitation quality, ensured consistent technical guidance and strengthened the institutional learning process. Follow-up actions include integration of digital systems into routine extension services, data validation, refresher training and expanded access to sustain learning, planning and institutionalization.

Implementation of work plan and budget

Activities were implemented largely on schedule, with minor initial delays mitigated through alternative arrangements. The project remained within the allocated budget. Risk management was actively conducted, with ongoing monitoring of potential risks such as seedling availability, weather fluctuations and input delivery, and contingency measures were applied to prevent major disruptions. Environmental and social risks were addressed through participatory monitoring and engagement with farmers, while external risks, including transport constraints, were managed to the extent possible.

Sustainability

1. Capacity development

The project was designed to support long-term sustainability through capacity development, alignment with policies, and integration into existing organizational structures. Training and technical support reinforced the adoption of CSA practices, and FFS methodology was embedded within MALR structures, with the Agricultural Research Center (ARC) positioned to provide ongoing technical support. Strong collaborations with the MALR, the ARC, non-governmental organizations and community associations enhanced institutional sustainability. Tools, processes and responsibilities were handed over to ensure continuity, with follow-up actions including replication and monitoring.

2. Gender equality

Participation was limited to male farmers due to prevailing cultural norms and the male-dominated nature of sugarcane production in the target areas. While women were not direct beneficiaries, the project respected local norms and focused on promoting equitable access to training, inputs and income opportunities for male participants.

3. Environmental sustainability

Environmental sustainability was a strong component of the project. Interventions promoted CSA, soil and water conservation, and efficient resource use. Farmers, local authorities and implementing partners received training and education to support lasting adoption of sustainable practices.

4. Human Rights-based Approach (HRBA) – in particular Right to Food and Decent Work

The project also reinforced human rights principles, including participation, accountability, non-discrimination, transparency, human dignity and empowerment, by involving male farmers in decision-making processes. It contributed to the Right to Food by improving access through increased productivity and income generation. Labour conditions were improved through training on safe and efficient agricultural practices, while climate-smart techniques reduced labour inefficiencies.

5. Technological sustainability

Technologies introduced were appropriate, context-specific and cost-effective. Raised-bed transplanting, irrigation scheduling, earthing up and IPM were suitable for adoption and maintenance by male farmers, with low environmental impact. The project strengthened local knowledge and technical capacity through FFS and ToT programmes. Beneficiaries demonstrated moderate capacity to sustain practices independently, with continued support from MALR and ARC likely to enhance long-term outcomes.

6. Economic sustainability

Economically, the project facilitated additional financial resources for the sugarcane sector by demonstrating the effectiveness of improved practices, attracting interest from government, donors and development partners. The technologies and practices introduced were affordable and feasible for farmers, although continued technical guidance and access to inputs are needed to ensure sustained adoption.

Achievement of results - Logical framework

<table>
<tr><td>Expected Impact</td><td colspan="2">Contributed to improving agricultural productivity of Sugarcane and managing natural resources (specifically water, land & fertilizers) through enhancing the technical capacity of Sugarcane smallholder producers in Egypt</td></tr>
<tr><td rowspan="5">Outcome</td><td colspan="2">Sugarcane productivity of smallholder farmers in upper Egypt enhanced with less irrigation and fertilizers through the adoption of climate-resilient good agriculture practices applying Field Farms Schools (FFSs) and demonstration fields</td></tr>
<tr><td>Indicator</td><td>Percentage of increase of sugarcane yield production.</td></tr>
<tr><td>Baseline</td><td>34 tonnes/feddan.</td></tr>
<tr><td>End Target</td><td>10 percent yield increase of smallholder sugarcane farmers.</td></tr>
<tr><td>Comments and follow-up action to be taken</td><td>The project demonstrated strong progress towards its intended outcomes, with clear improvements in sugarcane performance resulting from the introduced practices. Key interventions, including transplanted sugarcane on raised beds, timely and uniform earthing up, optimized irrigation scheduling, improved fertilization and strategic intercropping, were implemented and validated across 20 FFS demonstration fields. Field inspections and expert sampling showed that transplanted sugarcane produced seven to 15 tillers per plant, confirming the effectiveness of earthing up and related practices in enhancing basal tillering. This high tiller density indicates an increase in stalk numbers and yield potential, demonstrating the agronomic benefits of the promoted CSA practices.
Although final harvesting will occur after project closure, preliminary observations indicate substantial productivity gains. Estimated yields range from 45 to 65 tonnes per feddan, averaging approximately 55 tonnes per feddan, representing an increase of around 62 percent over the baseline. These results exceed end-of-project targets and provide strong evidence of impact at the demonstration level.
At the outcome level, sustainability depends on adoption and replication beyond FFS plots. While a full assessment requires a structured survey after one production cycle, qualitative evidence from focus group discussions indicates strong farmer willingness to apply the practices independently. The FFSs engaged 500 smallholder farmers managing 594 feddans, indicating significant outreach and potential for post-project diffusion.
Follow-up actions include a comprehensive post-project adoption and impact survey to quantify yield gains, assess resource-use efficiency and document farmer success stories, thereby supporting scaling up and sustained application of the CSA practices.</td></tr>
</table>

<table>
<tr><td rowspan="3">Output 1</td><td colspan="3">Training centers, attached to the newly established sugarcane transplants production centers (nurseries) in Edfu and Kom Ombo (one each), operationalized</td></tr>
<tr><td>Indicators</td><td>Target</td><td>Achieved</td></tr>
<tr><td>Number of operationalized training centres for the capacity building of sugarcane producers in Aswan in upper Egypt.</td><td>2</td><td>Partially</td></tr>
<tr><td>Baseline</td><td colspan="3">0</td></tr>
<tr><td>Comments</td><td colspan="3">At project closure, Output 1 was partially achieved. The Kom Ombo training centre was fully operational, whereas the planned Edfu centre could not be completed due to factors beyond the project’s control. The Kom Ombo centre, linked to a 40 feddan demonstration and seedling multiplication area, was rehabilitated and equipped as a multipurpose capacity development hub. It provides classroom training on transplanting, raised-bed cultivation, irrigation, fertilization and pest management, alongside practical field demonstrations aligned with the FFS methodology. The centre also facilitates training for facilitators, planning of FFS sessions, review of field observations and documentation of lessons learned. Farmer group discussions and participatory decision-support sessions are conducted on site, while data collection, reporting and monitoring are supported by information and communication technology (ICT) and office infrastructure. Over the course of the project, the centre evolved into a functional service-delivery unit capable of sustaining training activities beyond the project.
The Edfu centre in Wadi El-Sa‘idia remained under government construction and could not be rehabilitated. To mitigate this, a five feddan demonstration field was established in Al-Iman Village, incorporating onion intercropping and serving as a decentralized training platform for facilitators and technical staff. This field enabled hands-on training, field observation and comparative analysis, compensating for the absence of a permanent centre.
Follow-up actions include completion of construction and formal handover of the Edfu centre for future operationalization, integration of the centre into follow-on projects or national programmes to finalize equipment and staffing, and maintenance of demonstration fields as satellite training platforms. Coordination among the ARC, sugar factories and local authorities will be strengthened to embed the training centre operations within the national sugarcane development strategy.</td></tr>
</table>

Activity 1.1	Allocation of a suitable place for the two training centers	
	Achieved	Partially
	Comments	Suitable locations for two training centres were identified in coordination with the national counterpart at sugarcane seedling production stations in Kom Ombo and Edfu (Wadi El-Sa'idia). The Kom Ombo site was fully allocated, accessible and immediately available, allowing the successful establishment of an operational training centre. In contrast, although the Edfu site was technically identified and agreed upon, the facility remained under government construction and was not officially handed over during project implementation, preventing any rehabilitation works. Completion and formal handover of the Edfu facility were beyond the project's scope. Once these processes are finalized, the site may be considered for rehabilitation and operationalization under future projects or national programmes.
Activity 1.2	Identification of the needed training tools and equipment	
	Achieved	Yes
	Comments	The required training tools, furniture and equipment were comprehensively identified through consultations with ARC technical staff, nursery managers and FAO technical experts. The assessment considered the needs of classroom training, facilitator coordination, data management and farmer group learning, ensuring alignment with FFS methodology and adult learning principles. No major impediments were encountered during this process. The finalized list formed the technical basis for procurement and ensured that the equipment and materials were relevant both for immediate training delivery and for long-term institutional use.
Activity 1.3	Purchasing tools and equipment for the two training centers	
	Achieved	Yes
	Comments	Procurement of the identified tools and equipment was successfully completed in accordance with FAO procedures. All items intended for the Kom Ombo training centre were delivered, installed and made operational. These included ICT equipment, office furniture, training and meeting furniture, and storage units. While procurement covered the planned scope, installation at Edfu could not take place due to the unavailability of the physical facility. The procured equipment was therefore used to fully equip the Kom Ombo centre and to support decentralized training activities. Follow-up actions include allocating additional equipment to the Edfu centre once the facility becomes available, either through redeployment or complementary funding.
Activity 1.4	Functioning of the procured training tools and equipment in sites	
	Achieved	Yes
	Comments	The procured tools and equipment were effectively deployed and are fully functional at the Kom Ombo training centre. They supported a range of activities, including technical training sessions, facilitator and farmer meetings, reporting, monitoring and coordination with project partners. No significant operational constraints were observed. The availability of functional equipment enhanced the quality of training, improved documentation and strengthened institutional capacity at the local level. Follow-up actions include routine maintenance of equipment and continued use of the facilities for post-project training activities under national oversight.
Activity 1.5	Establishment of the training sugarcane field plots of the training centers	
	Achieved	Yes
	Comments	Training and demonstration plots were successfully established to support experiential learning. In Kom Ombo, a 40 feddan demonstration and seedling multiplication area was directly linked to the training centre, enabling continuous hands-on training and field-based learning. In Edfu, despite the absence of a physical training centre, a five feddan demonstration field in Wadi El-Sa'idia (Al-Iman Village) served as a practical training platform. The field showcased transplanted sugarcane intercropped with onion and reinforced key messages on productivity, system intensification and efficient use of resources. These demonstration plots effectively compensated for infrastructure limitations and ensured the continuity of capacity development activities. Follow-up actions include maintaining the plots as reference sites and expanding their use in future FFS cycles and extension programmes.

<table>
<tr><td rowspan="3">Output 2</td><td colspan="4">Capacities of national cadres (master trainers) in the field of sugarcane climate smart agriculture (CSA) for small farmers enhanced through training of trainers</td></tr>
<tr><td colspan="2">Indicators</td><td>Target</td><td>Achieved</td></tr>
<tr><td colspan="2">Number of trained national cadres under ToT (master trainers).</td><td>20 national cadres (master trainers).</td><td>Yes</td></tr>
<tr><td>Baseline</td><td colspan="4">0</td></tr>
<tr><td>Comments</td><td colspan="4">A total of 22 national cadres were trained as technical master trainers, exceeding the original target of 20. Based in Edfu and Kom Ombo, they actively supported the implementation of sugarcane FFSs. Participants were selected from key institutions, including extension directorates, the Sugar Crops Research Institute, the Faculty of Agriculture, technical secondary schools and sugar factories, thereby strengthening technical capacity and interinstitutional coordination. Three participants, representing 13.6 percent of the group, were female, reflecting the sectoral gender distribution, while equal participation opportunities were ensured in line with FAO principles.
The ToT programme, held from 1 to 4 September 2024 at the Pyramisa Hotel in Aswan, covered climate change impacts, CSA concepts, modern sugarcane cultivation including transplanting and raised-bed techniques, optimized irrigation and fertilization, natural resource management and FFS facilitation skills. Pre- and post-training assessments indicated substantial knowledge gains, with scores increasing from 45.6 percent to 99.4 percent.
Following the training, the master trainers supported 20 sugarcane FFSs in Edfu and Kom Ombo, providing technical guidance, contributing to awareness campaigns and reinforcing farmer confidence in adopting CSA practices. Follow-up actions include continuing the engagement of master trainers in future FFS cycles, providing refresher training, and integrating them into national extension and training programmes to ensure long-term sustainability.</td></tr>
<tr><td rowspan="3">Activity 2.1</td><td colspan="4">Developing a training programme for the ToT</td></tr>
<tr><td>Achieved</td><td colspan="3">Yes</td></tr>
<tr><td>Comments</td><td colspan="3">A structured and technically robust ToT programme was developed by FAO experts in close coordination with the national counterpart. The programme directly supported the project outcome by integrating CSA principles, modern sugarcane production practices, and the FFS methodology into a coherent learning framework. Training content was tailored to the agroecological conditions of Edfu and Kom Ombo, addressing water scarcity and the key production constraints in Upper Egypt. Particular emphasis was placed on practices requiring behavioural change, such as transplanting, raised-bed cultivation, earthing up and optimized irrigation, to equip trainers with the technical confidence needed to overcome farmer scepticism. No major impediments were encountered during programme development; however, limited availability of localized CSA materials required adaptation of existing FAO resources rather than the full development of new modules. Follow-up actions include updating the programme based on lessons learned from FFS implementation and converting the materials into standardized national training modules to support future replication.</td></tr>
<tr><td rowspan="3">Activity 2.2</td><td colspan="4">Identifying 20 males and females cadres for the Training of Trainers (ToT) for master trainers</td></tr>
<tr><td>Achieved</td><td colspan="3">Yes</td></tr>
<tr><td>Comments</td><td colspan="3">Master trainers were selected through a structured, criteria-based process that prioritized technical expertise in sugarcane production, experience in extension work, and institutional roles enabling sustained engagement with farmers. The final group of 22 trainers exceeded the project target and included representatives from extension services, the Sugar Crops Research Institute, sugar factories, universities and agricultural technical schools. This diversity strengthened cross-institutional ownership and reduced reliance on a single channel for disseminating CSA practices. Female participation was limited to 13.6 percent, reflecting the low representation of women within sugarcane-related institutions in Upper Egypt. While structural constraints affected gender balance, the selection process itself was inclusive and placed no restrictions on female participation. Follow-up actions include targeted capacity development for female extension staff and early identification of potential female trainers in future programmes to gradually improve gender balance.</td></tr>
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<table>
<tr><td rowspan="3">Activity 2.3</td><td colspan="4">Conducting the ToT for master trainers in the established training centers in Edfu and Kom Ombo</td></tr>
<tr><td>Achieved</td><td colspan="3">Yes</td></tr>
<tr><td>Comments</td><td colspan="3">The ToT programme was successfully implemented and led to substantial improvements in participant knowledge, as demonstrated by pre- and post-training assessments. The training combined technical lectures, interactive discussions and applied examples, enabling participants to internalize both technical content and FFS facilitation skills. Although initially planned for delivery through the training centres in Edfu and Kom Ombo, logistical considerations required centralized implementation in Aswan. This adjustment did not affect the quality of training but highlighted the need for flexible delivery modalities within project timelines. A key strength of the programme was its immediate linkage to field application, with trained master trainers directly supporting 20 FFSs, thereby reducing the gap between training and practice. However, the short duration limited opportunities for extended field-based practice and peer-to-peer mentoring. Follow-up actions include refresher training, on-the-job coaching during future FFS cycles, and systematic performance monitoring of master trainers to ensure sustained facilitation quality and technical effectiveness.</td></tr>
<tr><td rowspan="3">Output 3</td><td colspan="4">Capacities of national Field Farmers Schools (FFSs) facilitators in the field of sugarcane climate smart agriculture (CSA) for small farmers enhanced</td></tr>
<tr><td colspan="2">Indicators</td><td>Target</td><td>Achieved</td></tr>
<tr><td colspan="2">Number of trained FFSs facilitators.</td><td>20 facilitators.</td><td>Yes</td></tr>
<tr><td>Baseline</td><td colspan="4">0</td></tr>
<tr><td>Comments</td><td colspan="4">The project exceeded its target of 20 FFS facilitators, training 47 participants through two programmes in Aswan, held from 19 to 23 April and 17 to 21 August. Participants were drawn from extension services, the Sugar Crops Research Institute, sugar factories, graduates and related institutions in Edfu and Kom Ombo. The FAO-designed curriculum combined conceptual sessions with practical exercises, covering CSA, climate-smart water management, facilitator roles, community entry, AESA, FFS planning, adult learning, communication and operational planning. Field-based exercises in Sabil Village provided trainees with opportunities to practice AESA, facilitation, presentation and discussion management, building applied skills and confidence, while peer learning addressed differences in experience levels. Follow-up actions include continued mentoring during early FFS cycles, provision of refresher training, and systematic performance monitoring to ensure adherence to FFS principles.</td></tr>
<tr><td rowspan="3">Activity 3.1</td><td colspan="4">Developing draft CSA curricula for Sugarcane FFSs</td></tr>
<tr><td>Achieved</td><td colspan="3">Yes</td></tr>
<tr><td>Comments</td><td colspan="3">Draft curricula for CSA in sugarcane FFSs were developed in alignment with FAO FFS guidelines and adapted to the agroecological conditions of Upper Egypt. The curricula integrated climate change concepts, CSA practices, modern sugarcane production techniques and seasonal learning cycles suitable for farmer-led experimentation. While the curricula effectively covered core content, continuous refinement was required during implementation to respond to farmer feedback and field realities. Follow-up actions include consolidating the refined curricula into standardized national training manuals to support future replication and scale-up.</td></tr>
<tr><td rowspan="3">Activity 3.2</td><td colspan="4">Identification of trainers for the FFSs facilitators training</td></tr>
<tr><td>Achieved</td><td colspan="3">Yes</td></tr>
<tr><td>Comments</td><td colspan="3">Qualified trainers were selected from FAO technical experts, the Sugar Crops Research Institute and experienced FFS practitioners. Selection focused on technical expertise, familiarity with participatory extension methods, and experience in CSA and sugarcane systems. The main constraint was the limited pool of nationally certified FFS trainers with strong sugarcane specialization, which required FAO technical staff to take a leading role. Follow-up actions include gradually transitioning training responsibilities to national institutions by certifying experienced facilitators and master trainers as future trainers of facilitators.</td></tr>
<tr><td rowspan="3">Activity 3.3</td><td colspan="4">Delivering the training for FFSs facilitators</td></tr>
<tr><td>Achieved</td><td colspan="3">Yes</td></tr>
<tr><td>Comments</td><td colspan="3">The five day facilitator training was successfully delivered using a blended approach that combined classroom instruction, group exercises and field-based practice. The training effectively bridged the gap between technical knowledge and facilitation skills, which are critical for successful FFS implementation. A key strength of the programme was the inclusion of hands-on facilitation practice under real field conditions, enhancing participant confidence and practical competence. However, the relatively short duration limited opportunities for repeated facilitation practice. Follow-up actions include on-the-job coaching during FFS implementation, periodic refresher sessions, and structured peer learning among facilitators to sustain quality and consistency.</td></tr>
</table>

Output 4	Demonstration fields through the FFSs implemented in the main sugarcane production areas, using drip irrigation in Aswan Governorate		
	Indicators	Target	Achieved
	Number of established pilot fields through FFSs extension plots.	20 FFSs	Yes
Baseline	0		
Comments	A total of 20 FFS demonstration fields were established across Kom Ombo and Edfu, with ten fields in each location, engaging 500 smallholder sugarcane farmers, 25 per FFS, managing a total of 594 feddans. Each FFS functioned as a hands-on learning hub, where farmers practised CSA techniques under the guidance of trained facilitators. Key practices introduced included transplanted sugarcane on raised beds measuring 150 by 40 centimetres, which improved water infiltration, reduced waterlogging and enabled efficient irrigation, cutting irrigation time per field from 12 to two hours. Earthing up enhanced basal tillering, increasing stalk density, uniformity and yield, while the raised beds facilitated easier execution. Intercropping with onion, zucchini and legumes increased land-use efficiency, diversified income and reduced economic risk without compromising sugarcane productivity. Optimized irrigation scheduling conserved water, ensured timely soil moisture and prevented over-irrigation. Efficient fertilizer and micronutrient management, including zinc, iron and manganese, improved nutrient-use efficiency, stalk strength and sugar content. Pest and disease management, particularly for sugarcane borer prevention, reduced crop losses and improved plant health. Farmers reported immediate benefits, including easier field management, reduced labour and irrigation time, and improved crop establishment. These demonstration fields provided a practical, scalable model of climate-smart sugarcane production suitable for replication on smallholder farms. Summary of FFS villages, practices and participants:		

No.	District	Village	Longitude	Latitude	Special Practices	No. of Participants
1	Kom Ombo	Al-Atmour Bahary	32.975748	24.519760	Transplanted sugarcane, raised-bed cultivation (150×40 cm); earthing up; zucchini intercropping; opening beds after intercrop harvest; irrigation scheduling	25
2		Al-Ahmadia	32.965451	24.500552	Transplanted sugarcane, raised-bed cultivation; earthing up; irrigation management; fertilizer use efficiency	25
3		Al-Raghama East	32.956843	24.523135	Transplanted sugarcane, raised-bed cultivation; earthing up; improved irrigation and fertilizer management	25
4		Sabil Al-Nazara	32.919232	24.491845	Transplanted sugarcane, raised-bed cultivation; earthing up; sugarcane borer management; irrigation scheduling	25
5		Al-Abbasia	32.989704	24.435577	Transplanted sugarcane, raised-bed cultivation; earthing up	25
6		Al-Basaly	32.976028	24.464278	Transplanted sugarcane, raised-bed cultivation; earthing up; onion intercropping; sugarcane borer control	25
7		Sabil Mekki	32.910865	24.485884	Transplanted sugarcane, raised-bed cultivation; earthing up; preventive sugarcane borer control	25
8		Al-Atmour Mostagad	33.013639	24.521250	Transplanted sugarcane, raised-bed cultivation; earthing up; preventive sugarcane borer control	25
9		Hegaza	32.974278	24.490083	Transplanted sugarcane, raised-bed cultivation; earthing up; onion intercropping	25
10		Al-Menshia	32.999652	24.557415	Transplanted sugarcane, raised-bed cultivation; earthing up	25
11	Edfu	Al-Basiliya	32.787170	25.103592	Transplanted sugarcane, raised-bed cultivation; earthing up; sugarcane borer management; micronutrients (zinc, iron and manganese)	25
12		Al-Kelh	32.842918	25.052373	Transplanted sugarcane, raised-bed cultivation; earthing up; preventive sugarcane borer control; micronutrients (zinc, iron and manganese)	25
13		Al-Salayma	32.855120	24.992750	Transplanted sugarcane, raised-bed cultivation; earthing up	25
14		Al-Mareinab	32.859006	24.891215	Transplanted sugarcane, raised-bed cultivation; earthing up; micronutrients (zinc, iron and manganese) and humic substances	25
15		Hager Abu Khalifa	32.857651	24.966361	Transplanted sugarcane, raised-bed cultivation; earthing up	25
16		Sheikh Mahmoud	32.847865	24.982042	Transplanted sugarcane, raised-bed cultivation; earthing up; micronutrients (zinc, iron and manganese) and humic substances	25
17		Al-Fawza	32.894317	24.924447	Transplanted sugarcane, raised-bed cultivation; earthing up; onion intercropping	25
18		Al-Sa'ayda 1	32.789928	24.882042	Transplanted sugarcane, raised-bed cultivation; earthing up; onion intercropping	25
19		Al-Sa'ayda 2	32.782888	24.995723	Transplanted sugarcane, raised-bed cultivation; earthing up; onion intercropping	25
20		Al-Howeidiya	32.858858	24.936017	Transplanted sugarcane, raised-bed cultivation;	25

Output 4	Demonstration fields through the FFSs implemented in the main sugarcane production areas, using drip irrigation in Aswan Governorate			
	Indicators		Target	Achieved
	Number of established pilot fields through FFSs extension plots.		20 FFSs	Yes
Baseline	0			
Comments	The project initially planned to install solar-powered drip irrigation for all FFS demonstration fields to improve water-use efficiency and precise nutrient delivery. Although technical specifications were prepared and approved, procurement was not pursued due to anticipated delays that could have affected project timelines. The alternative approach using raised beds delivered several key benefits. Raised beds allowed farmers to irrigate each field in two hours compared with twelve hours on flat beds, reducing both labour and water use. They enabled timely and uniform earthing up, which promoted basal tillering and higher stalk density; installing drip lines first could have constrained this practice and reduced yield potential. Agronomically, raised beds reduced waterlogging, improved root aeration and nutrient uptake, simplified fertilizer and pesticide application, and enhanced labour efficiency and farmer acceptance. Farmers reported healthier plants, more uniform establishment and expressed strong willingness to adopt raised-bed transplanting on their own farms.			
Activity 4.1	Selection of around 300 -500 small sugarcane producer, participants of FFSs			
	Achieved	Yes		
	Comments	This activity was fully achieved, with 500 smallholder sugarcane farmers successfully selected and engaged in FFSs across Kom Ombo and Edfu. Farmer selection followed a transparent, inclusive and systematic process to ensure effective learning, equitable representation and high adoption potential of CSA practices. Priority was given to farmers with small to medium landholdings of less than three feddans to maximize benefit and replicability. Consideration was also given to commitment to adopt practices such as transplanted sugarcane, raised beds, improved fertilization and earthing up, as well as readiness to engage fully in field activities, discussions and collective decision making. The process included village consultations involving staff, facilitators and local leaders, nominations by village committees based on agreed criteria, field visits to verify landholding, cropping systems and commitment, and final enrolment of 500 farmers, ensuring a balanced mix of land sizes and experience for effective peer learning. Follow-up actions include tracking adoption of introduced practices and promoting farmer-to-farmer diffusion in neighbouring villages.		
Activity 4.2	Providing support to the FFS Facilitators to guide the implementation of 20 FFSs			
	Achieved	Yes		
	Comments	This activity was fully achieved, with continuous technical backstopping provided to all 20 sugarcane FFS facilitators in Kom Ombo and Edfu by FAO consultants, the national sugarcane expert and the project team, ensuring technical rigor and consistency. Facilitators received hands-on training and field coaching throughout the full transplanted sugarcane cycle, with particular emphasis on optimizing irrigation scheduling under raised beds to maintain soil moisture while reducing water losses, correct fertilizer and nutrient management including timing, rates, placement and use of micronutrients and humic substances to enhance nutrient-use efficiency and tillering, and timely and uniform transplanted sugarcane management with earthing up to promote basal tillering and plant anchorage coordinated with irrigation and fertilization. Guidance was also provided on intercropping practices, including selection, arrangement and management of intercrops to improve land-use efficiency, soil cover and additional income, and on pest and disease management through field scouting, identification of key pests, notably sugarcane borer and IPM strategies. Facilitators were trained in participatory supervision and adult learning approaches, fostering experiential learning, group facilitation and participatory analysis to actively engage farmers in assessing field performance. This technical support resulted in high-quality facilitation, consistent technical messaging and strong farmer engagement, with variations in facilitator experience mitigated through targeted coaching and peer learning. Follow-up actions include refresher training, ongoing mentoring during scaling up, and systematic documentation of facilitation best practices for replication in future sugarcane FFS programmes.		

Activity 4.3	Procurement of needed equipment for the demonstration fields through the FFSs	
	Achieved	Yes
	Comments	This activity was successfully implemented, with all 20 FFS demonstration fields adequately equipped to support the full cycle of transplanted sugarcane production under CSA principles. The project provided sugarcane transplants to all FFS sites to ensure uniform and timely establishment. Initial delays due to limited local availability were mitigated through alternative sourcing and coordinated delivery, ensuring that transplanting was completed within the planned agronomic window and synchronized with raised-bed preparation, irrigation and fertilization. Each FFS received essential equipment and inputs for transplanted and raised-bed systems, including row markers and measuring devices to standardize bed dimensions of 150 by 40 centimetres, implements for uniform earthing up to enhance basal tillering, and calibrated tools for accurate fertilizer and micronutrient application. Where minor delays occurred, locally available tools were strategically used to maintain continuity of transplanting, earthing up and nutrient management, ensuring uniform crop establishment and optimal yield potential. The timely provision of transplants and appropriate equipment strengthened hands-on learning, enhanced farmer confidence and demonstrated the technical feasibility of improved sugarcane management systems promoted through the FFSs. Follow-up actions include documenting equipment utilization efficiency, assessing durability under field conditions and incorporating farmer feedback to refine future procurement strategies for scaling up FFS-based sugarcane interventions.
Activity 4.4	Establishment of the procured equipment	
	Achieved	Yes
	Comments	This activity was fully achieved, with all procured equipment installed, tested and operational across the 20 FFS demonstration sites in Kom Ombo and Edfu. The equipment transformed each FFS into a functional learning laboratory, enabling continuous hands-on application of climate-smart sugarcane practices throughout the cropping cycle. The operational equipment supported farmers in practising core CSA interventions, including improved irrigation scheduling under raised beds, precise and timely fertilizer and micronutrient application, effective pest and disease monitoring, and systematic field measurements and performance tracking. All tools were calibrated and demonstrated in situ by facilitators, ensuring consistency in bed dimensions, transplant spacing, fertilizer rates and earthing up operations. This hands-on approach allowed farmers to observe and accurately replicate recommended practices, reinforcing learning through repeated application. The availability of functional equipment reduced measurement errors, improved agronomic accuracy and strengthened field observations. Facilitators and farmers could assess crop responses, such as tillering, vigour and uniformity, and directly link these outcomes to implemented practices, enhancing understanding of productivity, labour savings and resource-efficiency benefits of CSA practices. No major impediments were encountered. Follow-up actions include maintaining equipment, conducting refresher calibration sessions for facilitators, and documenting performance to guide future scaling and institutionalization of FFS-based sugarcane interventions.
Activity 4.5	Providing FFSs men and women participants with group-based packages of sugarcane CSA and NRM equipment, tools, seedlings, chemicals, etc	
	Achieved	Yes
	Comments	This activity was fully implemented, with all FFS groups receiving sugarcane CSA and natural resource management inputs aligned with transplanted raised-bed systems to support hands-on learning. Each group received transplants, fertilizers, micronutrients including zinc, iron and manganese, humic substances and essential tools for bed preparation, transplanting, earthing up and field management, enabling immediate application and reinforcing technical skills and farmer confidence. Group-based distribution promoted peer learning, equitable access and proper input use under facilitator supervision. Farmers practised efficient irrigation, precise nutrient application and IPM, gaining practical experience while reducing environmental impact. At the completion of the FFS cycle, graduation ceremonies recognised farmers' progress, and 500 knapsack sprayers were distributed to support continued precision management. Additionally, 217 000 sugarcane seedlings were provided to facilitate adoption of the seedling system and improve productivity. Follow-up actions include monitoring continued use of inputs and equipment, assessing their impact on efficiency and documenting farmer-led replication to guide future programming.

<table>
<tr><td rowspan="3">Activity 4.6</td><td colspan="3">Organizing 20 field days for small producers (10 in Edfu and 10 in Kom Ombo), along the production cycle</td></tr>
<tr><td>Achieved</td><td colspan="2">Yes</td></tr>
<tr><td>Comments</td><td colspan="2">This activity was fully achieved, with 20 structured field days held across Kom Ombo and Edfu, ten in each location, timed to coincide with key sugarcane growth and management stages. Field days covered planting and transplanting, earthing up, intercrop harvest and irrigation monitoring, allowing farmers to compare traditional and CSA practices in real time. Farmers actively participated in hands-on demonstrations, observations, discussions and problem solving, strengthening analytical skills, decision making and confidence in innovations such as raised-bed transplanting, optimized fertilization and IPM. Peer-to-peer knowledge exchange further reinforced learning. Follow-up actions include monitoring adoption of practices, conducting post-harvest yield and economic assessments, documenting lessons learned and success stories and maintaining ongoing technical support through facilitators and FFS networks.</td></tr>
<tr><td rowspan="3">Output 5</td><td colspan="3">Enabling environment set to allow FFSs effective monitoring, evaluation and networking to ensure sustainability and continued learning process for smallholder farmers in the selected sites</td></tr>
<tr><td>Indicators</td><td>Target</td><td>Achieved</td></tr>
<tr><td>Number of FFSs effectively monitored, evaluated and networked.</td><td>20</td><td>Yes</td></tr>
<tr><td>Baseline</td><td colspan="3">0</td></tr>
<tr><td>Comments</td><td colspan="3">This output was fully achieved, with all 20 FFSs supported by a structured participatory monitoring, evaluation, and networking framework that strengthened learning and ensured sustainability beyond the project. A multidisciplinary backstopping team, including the project coordinator, field coordinators, sugarcane expert, FFS specialist and programme/monitoring and evaluation specialist, provided continuous support to facilitators, emphasizing learning-by-doing and collective reflection rather than compliance-based supervision. Additionally, FFS diaries documented meeting dates, technical topics, challenges and solutions, group decisions, field demonstrations, lessons learned and stakeholder visits, enabling tracking of progress, pattern recognition and critical reflection. Continuous coaching and mentoring by master trainers improved facilitation skills, technical accuracy, participatory methods and data recording, boosting facilitator confidence and performance. Standardized digital registers allowed tracking of session dates, attendance, implemented practices, farmer observations, crop management decisions, and fertilizer and irrigation use, supporting comparative analysis and cross-site learning. A WhatsApp network, the Aswan Sugarcane Expert Network, provided a real-time knowledge exchange platform for facilitators, trainers, experts and project staff to share observations, discuss technical issues, exchange solutions, conduct quick polls and disseminate advice, enhancing peer learning, rapid problem solving and collective ownership.</td></tr>
<tr><td rowspan="3">Activity 5.1</td><td colspan="3">Support the FFS facilitators to monitor and evaluate the farmers’ performance</td></tr>
<tr><td>Achieved</td><td colspan="2">Yes</td></tr>
<tr><td>Comments</td><td colspan="2">This activity was successfully implemented, with FFS facilitators receiving structured training and continuous technical backstopping on participatory monitoring and evaluation. Facilitators used diaries, observation checklists, reflection sessions and learning plots to systematically track farmers’ performance, focusing on behavioural change, decision making and quality of practice adoption, including transplanting, raised-bed preparation, earthing up, irrigation, nutrient and pest management. Regular reflection enabled farmers to compare conventional and CSA practices, critically evaluate outcomes and adapt management decisions. Facilitators documented observations, discussions, constraints and actions, strengthening evidence-based learning, farmer analytical skills, confidence, collective problem solving and consistency in practice adoption. Variations in facilitator experience affected data quality and peak agricultural periods sometimes limited reflection time. Follow-up actions include continued master trainer coaching, provision of simplified digital tools and refresher sessions to harmonize participatory monitoring and evaluation quality and support long-term institutionalization.</td></tr>
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<tr><td rowspan="3">Activity 5.2</td><td colspan="2">Backstop and monitor the FFSs facilitators to improve their performance</td></tr>
<tr><td>Achieved</td><td>Yes</td></tr>
<tr><td>Comments</td><td>This activity was fully achieved through structured backstopping by master trainers, FAO consultants and the national project team, combining on-site visits with remote technical support. Backstopping strengthened facilitators’ competencies in participatory facilitation, technical accuracy, session planning and data recording, emphasizing FAO FFS principles such as farmer-led learning, inclusive participation and critical analysis. As a result, facilitation quality became consistent across 20 FFSs, with clearer sessions, accurate agronomic guidance and complete monitoring records, boosting facilitator confidence and reducing variability. Limited in-person visits during peak periods and varied facilitator skills increased coaching demands. Follow-up actions include continued mentoring, periodic performance reviews, development of facilitation checklists and short refresher trainings to institutionalize quality assurance and ensure sustainability.</td></tr>
<tr><td rowspan="3">Activity 5.3</td><td colspan="2">Create and maintain two sites (nursery-level) databases of FFSs implemented</td></tr>
<tr><td>Achieved</td><td>Yes</td></tr>
<tr><td>Comments</td><td>This activity was successfully achieved with the establishment of digital nursery-level databases at Kom Ombo and Edfu, covering all FFSs. Standardized registers captured session dates, practices, farmer participation disaggregated by gender, management decisions, input and irrigation use, crop performance, yield and farmer feedback, enabling real-time aggregation, cross-site analysis and evidence-based reporting. The system improved monitoring and evaluation efficiency, ensured data consistency, reduced reliance on paper records, and serves as an institutional memory tool for learning continuity, troubleshooting and replication. Variations in digital literacy and intermittent connectivity affected timeliness, which was mitigated through coaching, simplified formats and offline data uploads. Follow-up actions include integrating the databases into routine extension systems, conducting data validation and refresher training and expanding access to enhance learning, planning and long-term impact.</td></tr>
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Partnerships and Outreach
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